DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING						
Course Code: 24CS330		***Course Title*** ***Data Structures***				
Total Credits: 04 ***L-60: T-0: P-0: SL and TW-60***		***SDGs addressed: 4, 9, 11***		***Course Category: PCC***	***Course Type: Theory***	
Teaching Learning Process		***Total Contact Hours/semester***	***Assessment Weightage and Marks***			
	Lecture	**60**		**CIE**	**SEE**	**Total**
	Tutorial	**-**	***Weightage, %***	***40***	***60***	***100***
	Practical	**-**	***Maximum Marks***	***40***	***60***	***100***
	Self-learning and Term Work	**60**	***Minimum Marks***	***20***	***25***	***45****
	Total	**120**				

Note: *For passing the student has to score a minimum of 45 Marks (CIE+SEE: 20 + 25 or 21 + 24)

Course Prerequisite: C programming concepts.

Course Objectives:

Sl. No.	Course Objectives
1	Provide the knowledge of basic data structures and their implementations.
2	Introduce how arrays, linked structures, stacks, queues, trees, and graphs are represented in memory and used by algorithms in problem solving.
3	Impart the benefits of dynamic and static data structures.
4	Understand the importance of data structures in the context of writing efficient programs.

Course Outcomes (COs)

CO	Course Outcomes	Highest Level of Cognitive Domain	Working Knowledge
CO1	Using the concept of pointers to arrays, structures, and functions, perform different operations on linear list with array storage representation.	L3	WK2,WK3
CO2	Use the concept of linked list representation to implement various operations on Linear lists with different types of linked list.	L3	WK2,WK4
CO3	Solve the problems of conversion and evaluation of different types of expressions using stack data structure, comprehend the implementation of different types of Queues.	L3	WK2,WK4
CO4	Using tree data structure solve the problems of searching, sorting, expression conversion and expression evaluation.	L3	WK2,WK4
CO5	Applying graph traversal techniques solve the problems of testing connectivity and acyclicity in a graph.	L3	WK2,WK4

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

Course Content / Syllabus:

Unit No.	Content	Lecture Hours
1	**Introduction to Data structures:** Basic Terminology, Elementary Data Structure Organization, Classification of Data Structures, Operations on Data Structures, Abstract Data Type. **Pointers:** Declaring Pointer Variables, Pointer Expressions and Pointer Arithmetic Null Pointers, Generic Pointers, Pointer to Pointers, Drawback of Pointers. **Arrays:** Pointers and Arrays, Arrays of Pointers, Pointers and Two-dimensional Arrays. **Structures and Unions:** Introduction, Nested Structures, Arrays of Structures, Structures and Functions, Self-referential Structures. **Self-Learning and Term Work*** • Multi-dimensional Arrays. • Strings. • Unions	12
2	**Linked Lists:** Introduction, Singly Linked Lists, Circular Linked Lists, Doubly Linked Lists, Circular Doubly Linked Lists, Header Linked Lists **Self-Learning and Term Work*** • Multi-linked Lists. • Applications of Linked Lists.	12
3	**Stacks:** Introduction to Stacks, Array Representation of Stacks , Operations on a Stack, Linked Representation of Stacks, Operations on a Linked Stack, Applications of Stacks, **Queues:** Introduction to Queues, Array Representation of Queues, Linked Representation of Queues ,Types of Queues **Self-Learning and Term Work*** • Multiple Stacks • Multiple Queues. • Applications of Queues	12
4	**Trees:** Introduction, Types of Trees, Creating a Binary Tree from a General Tree, Traversing a Binary Tree **Efficient Binary Trees:** Binary Search Trees, Operations on Binary Search Tree, Binary Expression Trees, Threaded Binary Trees **Self-Learning and Term Work*** • Applications of Trees.	12
5	**Graphs:** Introduction, Graph Terminology, Directed Graphs, Bi-connected Components, Representation of Graphs, Graph Traversal Algorithms. **Self-Learning and Term Work*** • Files and Their Organization	12

*** NOTE: Self-Learning and Term Work are evaluated only through continuous internal evaluation (CIE) and will not be assessed during the Semester End Examination (SEE).**

Text Books:

Sl. No	Author/s	Title	Publisher Details
1	Reema Thareja	Data Structures using C	2nd Edition, 2018, Oxford University Press.
2	Richard F.Gilberg and Behronz A. Forouzan	Data structures, A Pseudo code Approach with C	3rd Edition, Cengage learning, 2013

Reference Books:

Sl. No.	Author/s	Title	Publisher Details
1	Yedidyah, Augenstein, Tannenbaum	Data Structures Using C and C++	2nd Edition, PHI.
2	E Balaguruswamy	Data Structures Using C	McGraw Hill Education, Paperback 2017.
3	Ellis Horowitz and Sartaj Sahni	Fundamentals of Data Structures in C	2nd edition, Universities Press, 2014.
4	Seymour Lipschutz	Data Structures Schaum's Outlines	Revised 1st Edition, McGraw Hill, 2014.

Web Resources:

Sl. No.	Web Links
1	https://nptel.ac.in/courses/106102064/
2	https://nptel.ac.in/courses/106105085/

Course Articulation Matrix:

Course Outcomes	Program Outcomes											Program Specific Outcomes		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	3	-	-	-	-	-	-	-	-	3	-	3
CO2	3	3	3	3	-	-	-	-	-	-	-	3	-	3
CO3	3	3	3	3	-	-	-	-	-	-	-	3	-	3
CO4	3	3	3	3	-	-	-	-	-	-	-	3	-	3
CO5	3	3	3	-	-	-	-	-	-	-	-	3	-	3

High – 3, Medium – 2, Low – 1